Christian_Herrera_HW11

April 21, 2026

1 Homework 11.Data Visualization (Questions)

2 Assignment Submission Guidelines

Please follow the guidelines below for submitting your assignment:

1. **Submission Deadline:**
   - All assignments must be submitted **no later than 23:59 PM next Tuesday (04/21)**.
   - Late submissions will not be accepted.
2. **Complete Your Work:**
   - Finish your work in Google Colab and ensure that your notebook is saved.
3. **Submit Your Assignment on Canvas:**
   - Log in to **Canvas** and navigate to the assignment submission page.
4. **File Naming Convention:**
   - Please name your files as follows: `Lastname_Firstname_AssignmentName`
   - Example: `Alex_John_HW11.ipynb` and `Alex_John_HW11.pdf`..
5. **Technical Issues:**
   - If you encounter any technical issues with Canvas or your submission, please contact the TAs immediately **before the deadline** to avoid penalties.

3 Questions

3.0.1 Use the `housing_california.csv` dataset to answer the questions.

3.0.2 Question 1

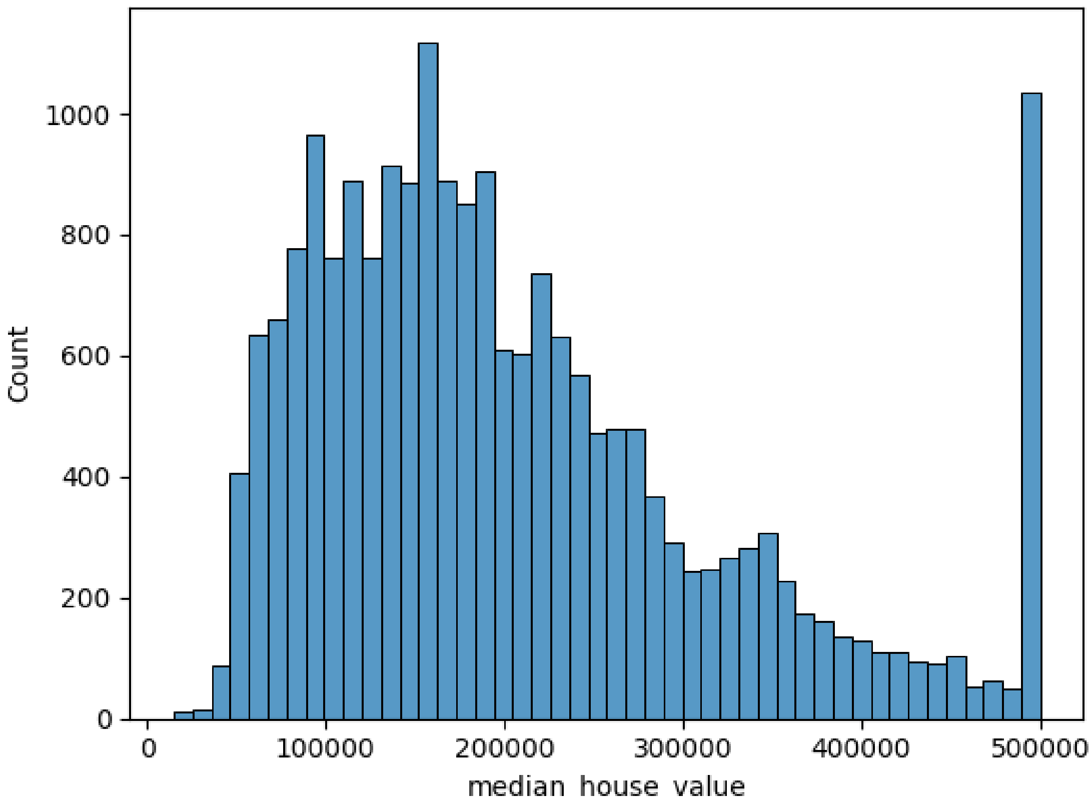
Histogram of Median Housing Prices:

[68]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

[69]:
```
df = pd.read_csv('housing_california.csv')

display(df.head())

sns.histplot(data=df, x='median_house_value')
plt.show()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	\
0	-122.23	37.88	41.0	880.0	129.0	
1	-122.22	37.86	21.0	7099.0	1106.0	
2	-122.24	37.85	52.0	1467.0	190.0	
3	-122.25	37.85	52.0	1274.0	235.0	
4	-122.25	37.85	52.0	1627.0	280.0	

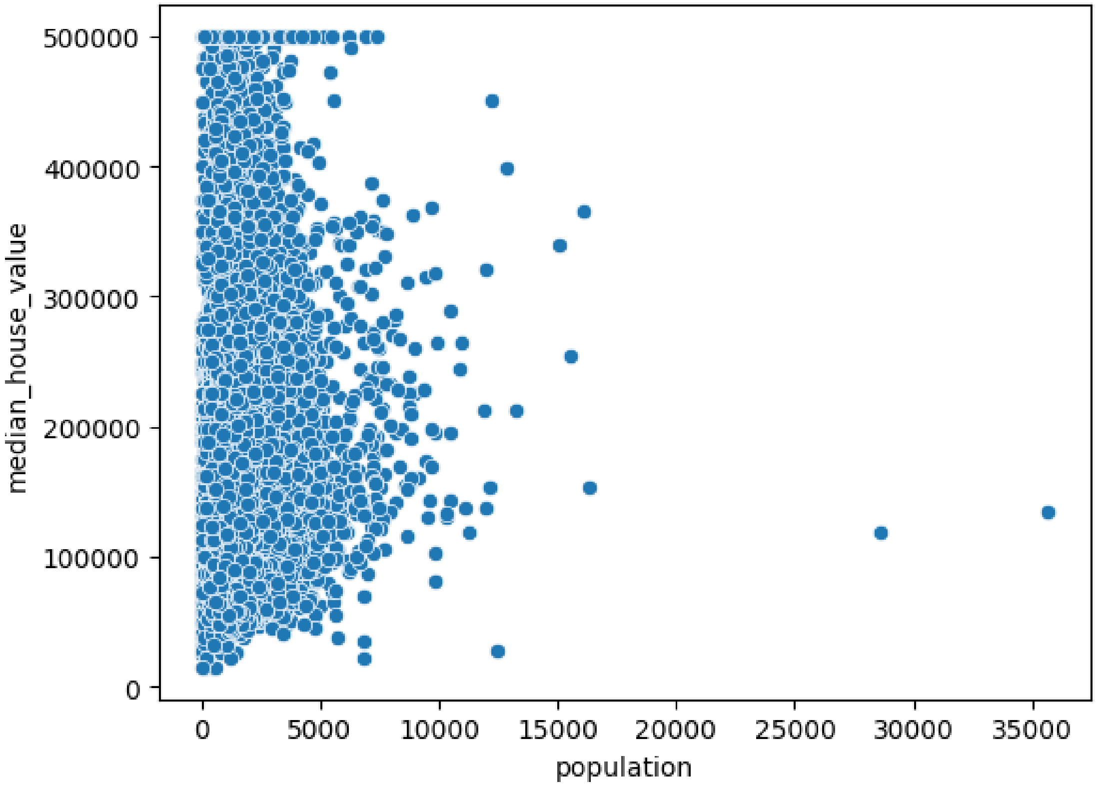
	population	households	median_income	median_house_value	ocean_proximity
0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	496.0	177.0	7.2574	352100.0	NEAR BAY
3	558.0	219.0	5.6431	341300.0	NEAR BAY
4	565.0	259.0	3.8462	342200.0	NEAR BAY



3.0.3 Question 2

Scatter Plot of Population vs. Median House Value:

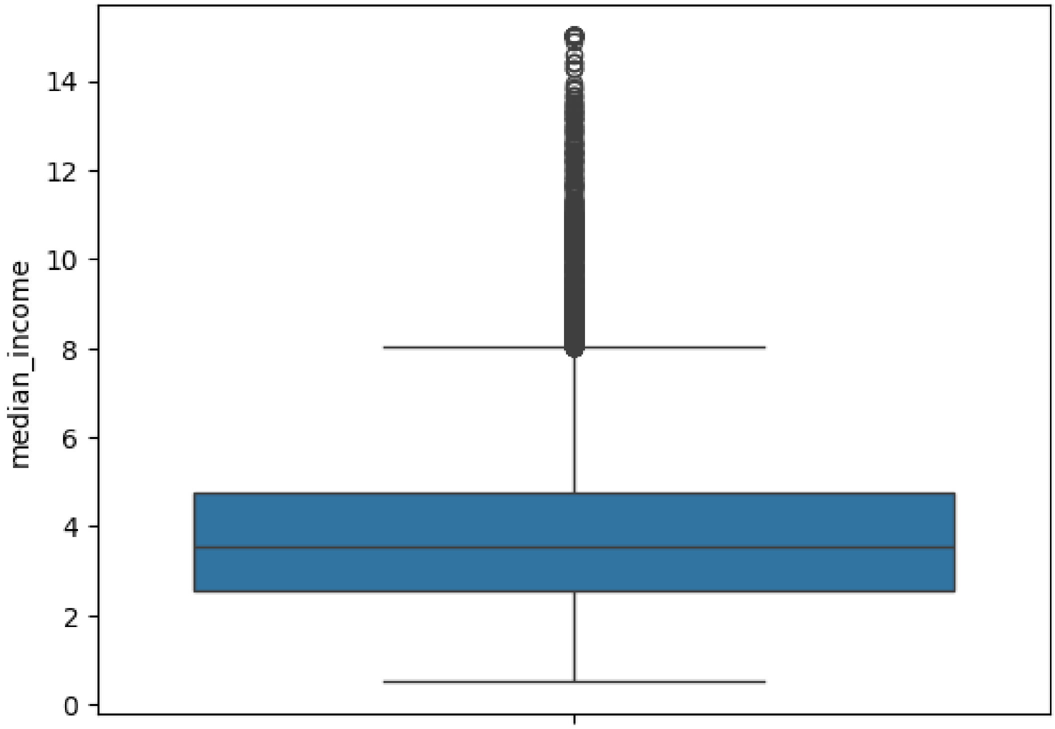
[70]:
```
sns.scatterplot(data=df, x='population', y='median_house_value')
# Not 100% sure that this is the correct orientation for the plot.
plt.show()
```



3.0.4 Question 3

Box Plot of Median Income:

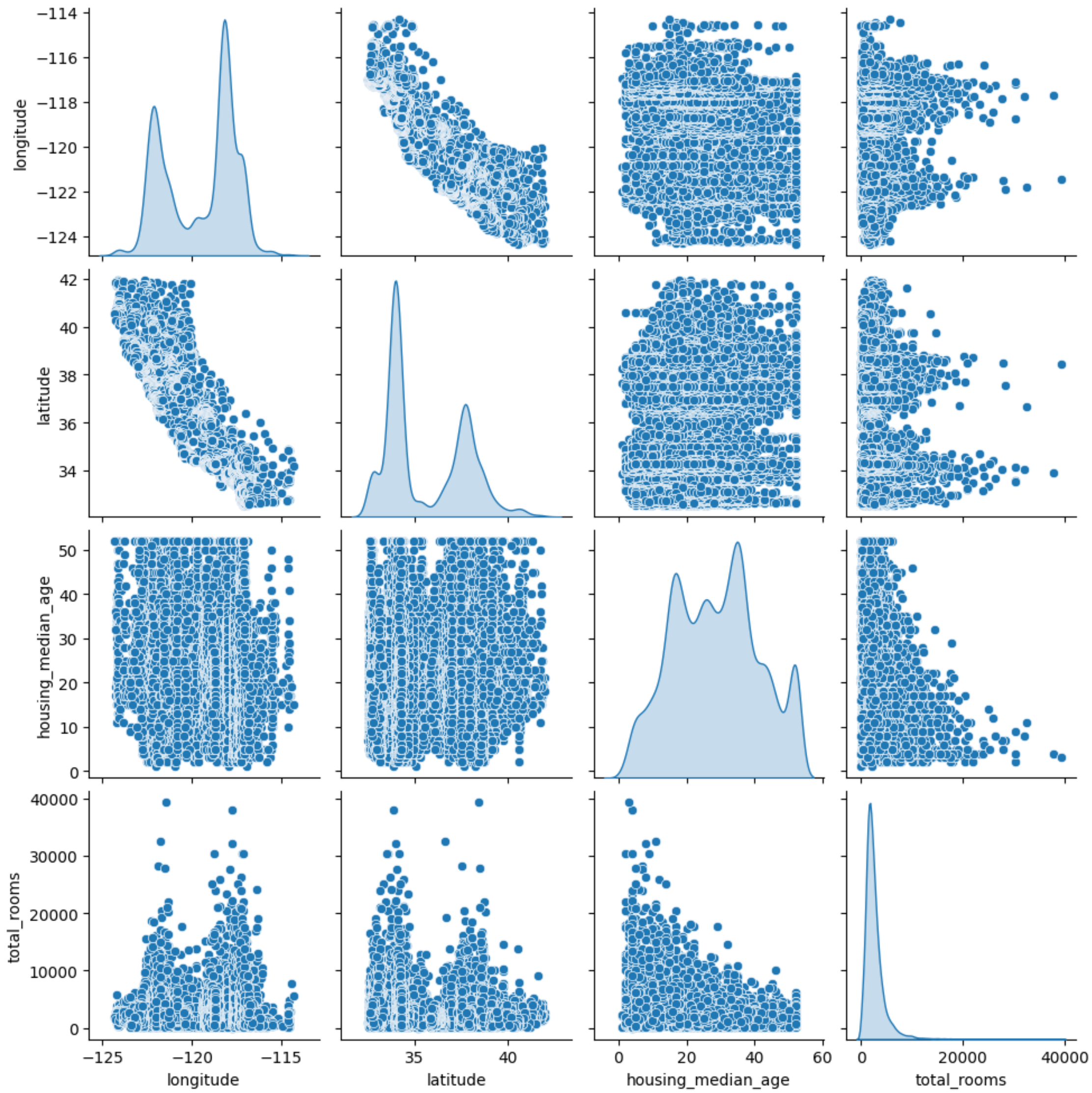
[71]:
```
sns.boxplot(data=df, y='median_income')
plt.show()
```



3.0.5 Question 4

Pair Plot for Select Columns:

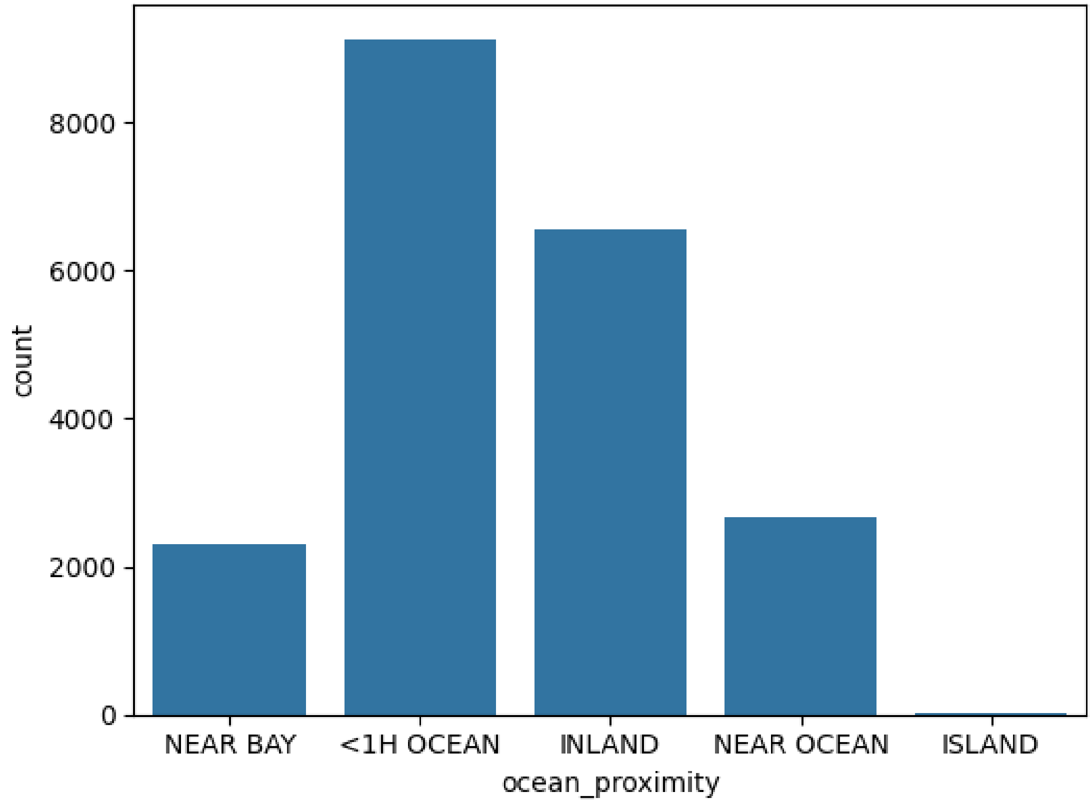
[72]:
```
# I used iloc to only the the pairplots of the first few columns, but with all␣
↪rows included.
sns.pairplot(data=pd.DataFrame(df.iloc[:,:4]), diag_kind = 'kde') # KDE␣
↪diag_kind for fun.
plt.show()
```



3.0.6 Question 5

Bar Plot of Ocean Proximity Counts:

[73]:
```
sns.countplot(data=df, x='ocean_proximity')
plt.show()
```

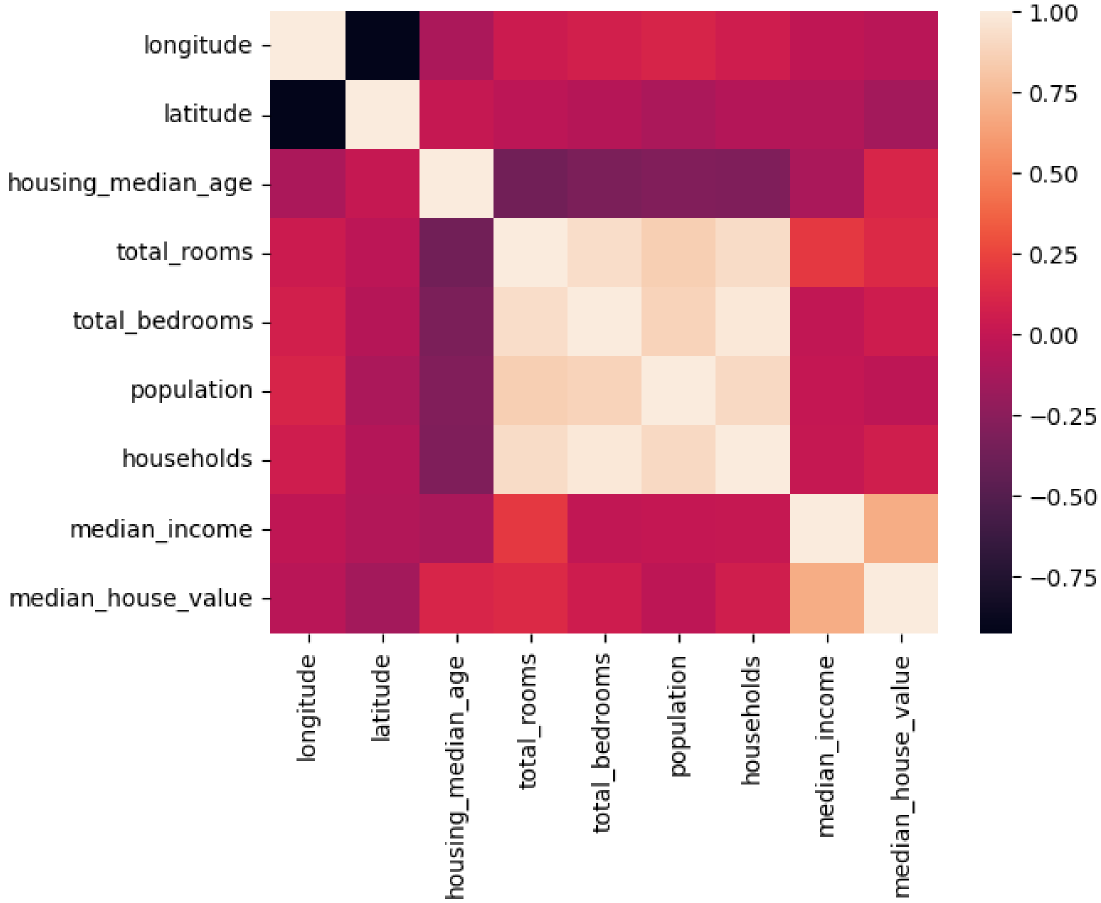


3.0.7 Question 6

Correlation Heatmap:

[74]:
```
sns.heatmap(df.corr(numeric_only=True))
```

[74]: <Axes: >

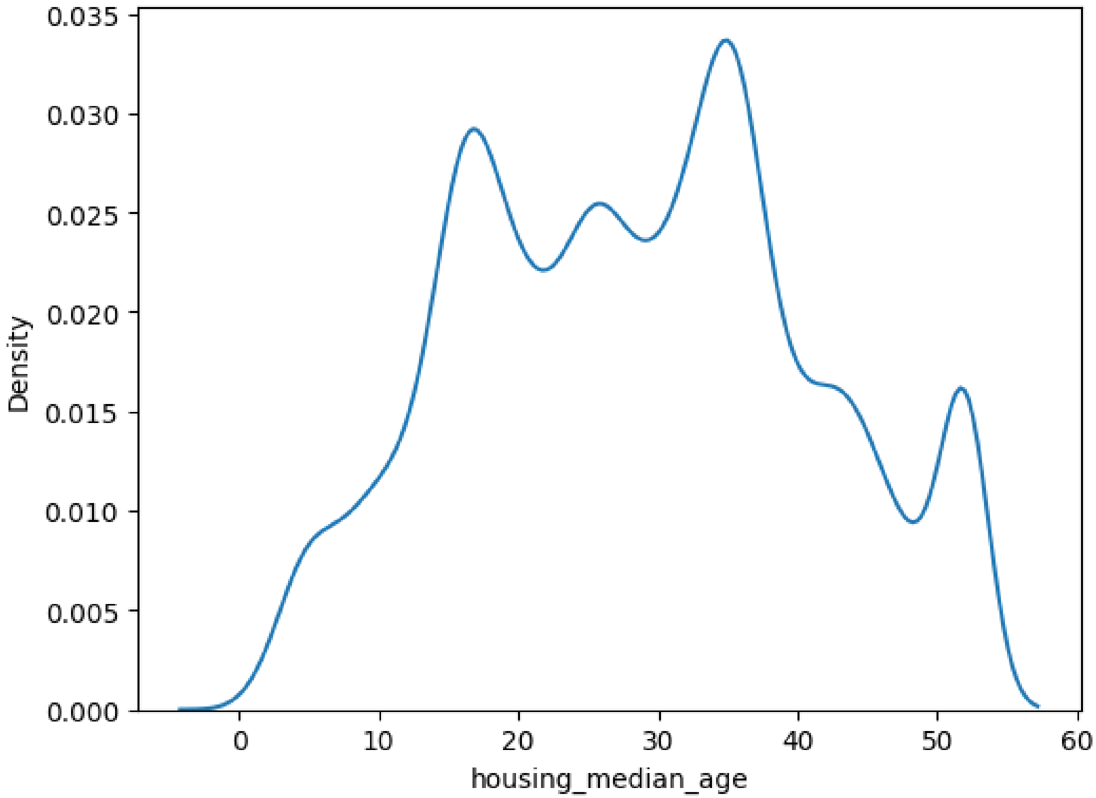


3.0.8 Question 7

Density Plot of Housing Median Age:

[75]:
```
sns.kdeplot(data=df, x='housing_median_age')
```

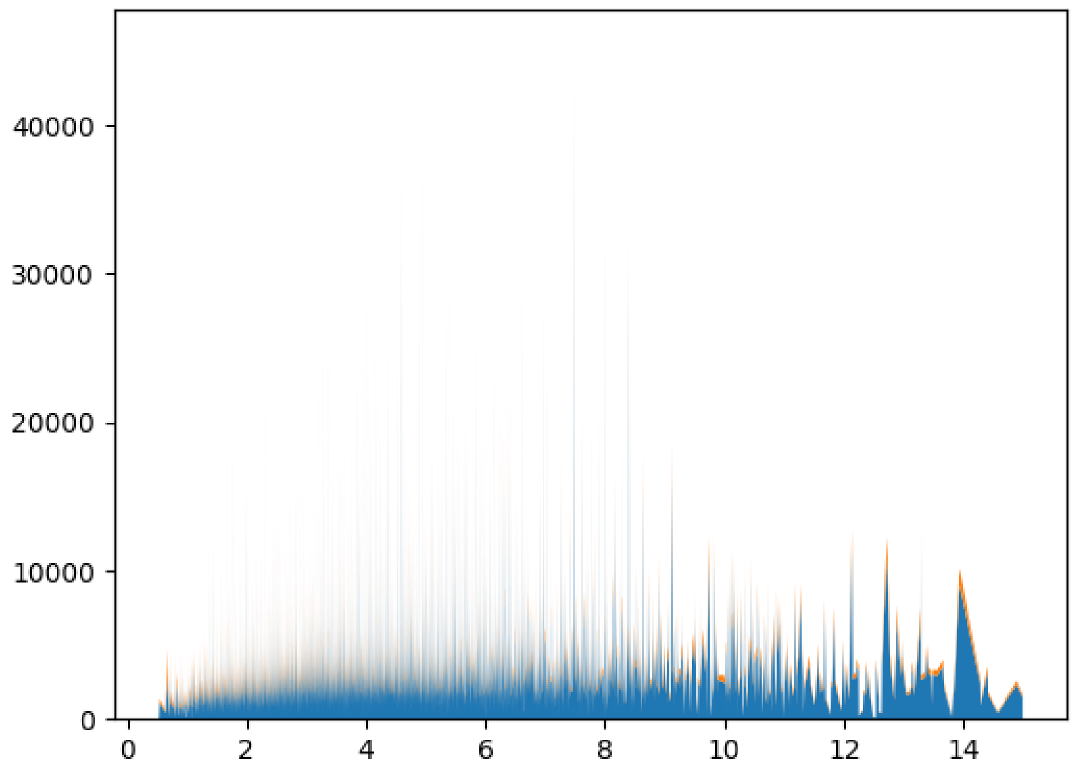
[75]: <Axes: xlabel='housing_median_age', ylabel='Density'>



3.0.9 Question 8

Stacked Area Plot of Total Rooms and Bedrooms:

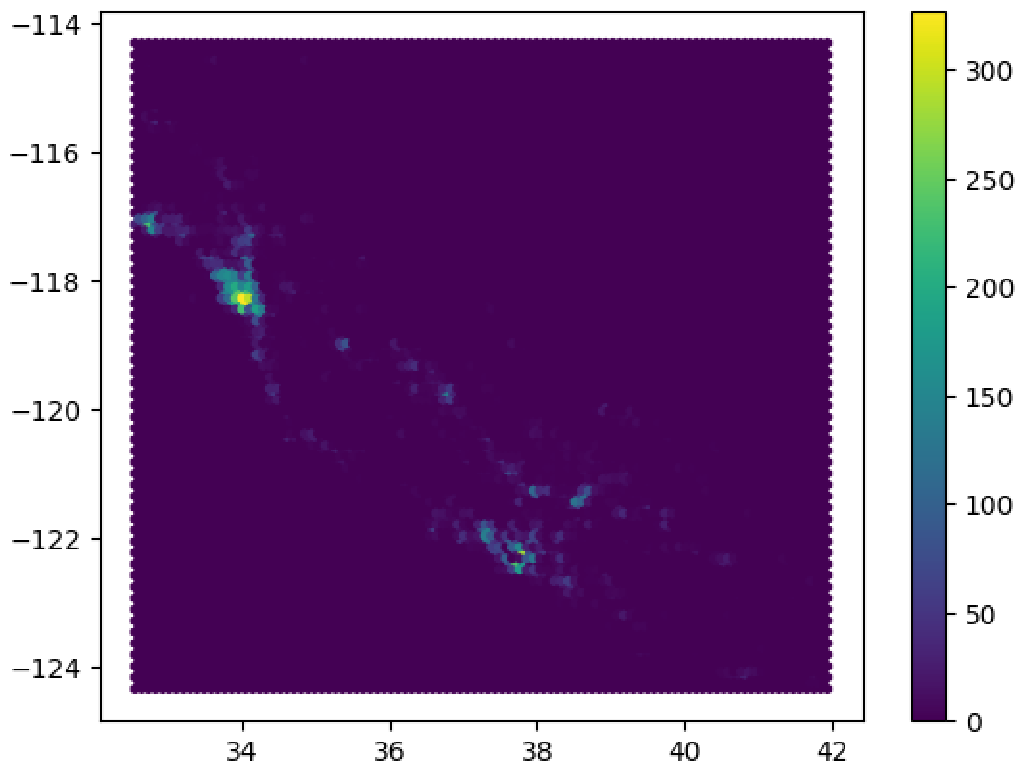
[76]:
```
df_sorted = df.sort_values('median_income')

plt.stackplot(
    df_sorted['median_income'],
    df_sorted['total_rooms'],
    df_sorted['total_bedrooms'])
plt.show()
```



3.0.10 Question 9

Hexbin Plot of Latitude and Longitude:

[77]:
```
plt.hexbin(x=df['latitude'], y=df['longitude'])
plt.colorbar()
plt.show()
```



3.0.11 Question 10

Facet Grid of Scatter Plots:

[78]:
```python
g = sns.FacetGrid(data=df, col='ocean_proximity')
g.map(sns.scatterplot, 'median_income', 'median_house_value')
plt.show()
```